

Autonomous Lane Following Vehicle

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github.com/UShouldRun/Autonomous-Driving

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Introduction

Motivation and project objective

Teaching a Car to See the Line

We train an autonomous vehicle with Reinforcement Learning to stay centred on a yellow lane line and avoid static obstacles, end to end, directly from raw sensor input.

The agent never receives a map or a rule set, only a camera frame, a LiDAR scan and its own speed at every step. Everything it does (steer, accelerate, swerve) is **learned** from a single shaped reward signal.

Webots 2023b simulator

Gymnasium step / reset API

Camera + LiDAR + speed

DQN vs. PPO comparison



Approach

Observation space, LaneCNN architecture, DQN & PPO

What the Agent Perceives

Camera

- $84 \times 84 \times 3$ RGB frame
- 115° FOV
- Tilted $\sim 6^\circ$ downward
- Bias toward the yellow line

LiDAR

- 8 vertical layers
- 115° horizontal FOV
- 23° vertical FOV clipped at $30m$
- Flattened into a norm. 1-D vector

State

- Current forward velocity
- Normalised to $[0, 1]$

Why fuse camera and LiDAR?

- Camera-only systems struggle give no direct distance signal
- LiDAR-only systems lack context such as lane colour.
- Fusing both streams: agent localises the line visually while reacting to obstacle proximity numerically.

LaneCNN

Camera branch

- 5 strided convolutional layers with batch-norm + ReLU
- AdaptiveAvgPool2d(4×4)

Layer	Channels	Kernel / Stride
Conv 1	$3 \rightarrow 32$	$3 / 1$
Conv 2	$32 \rightarrow 64$	$3 / 2$
Conv 3	$64 \rightarrow 128$	$3 / 2$
Conv 4	$128 \rightarrow 128$	$3 / 2$
Conv 5	$128 \rightarrow 256$	$3 / 2$
Pool	—	Adaptive 4×4

LiDAR branch

Two-layer MLP (LiDAR-dim $\rightarrow 64 \rightarrow 64$) with ReLU encodes proximity into a **64-d** vector.

State branch

Single linear layer maps the velocity scalar to a **16-d** embedding.

Joint head

The three embeddings are concatenated to form a $(512 + 64 + 16) = 592$ -d feature vector fed to the policy and value heads.

Algorithms — DQN vs. PPO

DQN — Discrete action space

Five named actions, each paired with a fixed forward throttle to prevent stalling:

Hard-Left

Hard-Right

Gentle-Left

Gentle-Right

Straight

Implemented with **experience replay** ($|\mathcal{D}| = 10^4$) and ε -greedy exploration.

Fixed actions simplify early exploration but limit fine steering resolution.

PPO — Continuous action space

Outputs steering $\varphi \in [-1, 1]$ and throttle $a \in [0, 1]$ as a diagonal Gaussian distribution.

Hyperparameter	Value
Clip ε	0.2
GAE- λ	0.95
Entropy coefficient	0.01
Learning rate	10^{-4}
Batch size	128

Reward & Evaluation

Reward shaping and the metrics used to score agents

Dense Reward Function

Reward decomposition

$$R = r_{\text{exist}} + r_{\text{progress}} + r_{\text{speed}} + r_{\text{lap}} - r_{\text{align}} - r_{\text{near}}$$

Term	Formula
Existence	$r_{\text{exist}} = w_e$
Progress	$r_{\text{progress}} = w_p \cdot \Delta d \cdot (1 + w_b(1 - \theta))$
Speed	$r_{\text{speed}} = w_s \cdot \hat{v} \cdot (1 + w_b(1 - \theta) + w_i \cdot \Delta \theta ^+)$
Alignment penalty	$r_{\text{align}} = w_a \theta $
Near-miss	$r_{\text{near}} = w_n \cdot \mathbb{1}[\text{near miss}]$
Lap bonus	$r_{\text{lap}} = w_{\text{lap}}$

Weight	Value
w_e (existence)	0.01
w_p (progress)	4.0
w_s (speed)	4.0
w_b (alignment)	4.0
w_i (improve)	10.0
w_a (penalty)	10.0
w_n (near miss)	5.0
w_{lap} (lap)	50.0
w_t (terminated)	100.0

Evaluation Metrics

| Success Rate (%)

Percentage of evaluation episodes completed without any collision or line-loss termination event.

Higher is better.

| Cross-Track Error — CTE (m)

Average lateral distance between the vehicle centre and the yellow centre line, measured every step.

Lower is better.

| Mean Lap Time (s)

Wall-clock time to complete one full circuit of the track. Rewards faster, smoother driving.

Lower is better.

| Near Miss count

Mean count of instances where the vehicle gets dangerously close to an obstacle recorded by LiDAR.

Lower is better.

Each model is evaluated over **10 episodes per track**. All metrics are averaged across those episodes.

Training Environments

Custom Webots tracks built for training and testing

Three Custom Looped Tracks

Track 1: Training

Simple rounded loop. Used for initial DQN and PPO training via **config.yaml**. Gentle curves only.



Track 2: Fine-tuning

Tighter curves and varying road width. Both agents fine-tuned here via **finetune.yaml**.



Track 3: Zero-shot eval

More complex layout: similar, but different curves. Used **exclusively** for zero-shot generalisation evaluation.



Experiments & Results

Action space · Cross-track generalisation · Obstacle avoidance

Experiment 1: Action Space

Goal

Compare DQN (discrete) and PPO (continuous) trained under the **same dense reward** on Track 1. Evaluate which action-space design produces better lane-keeping and lap efficiency.

Hypothesis

PPO will achieve smoother control due to continuous steering and throttle authority.
DQN's fixed action set may produce jagged corrections that keep the vehicle close to the line but slow it down.

Training setup

Environment:

Webots Track 1: simple rounded loop

Config:

config.yaml: shared hyperparameters

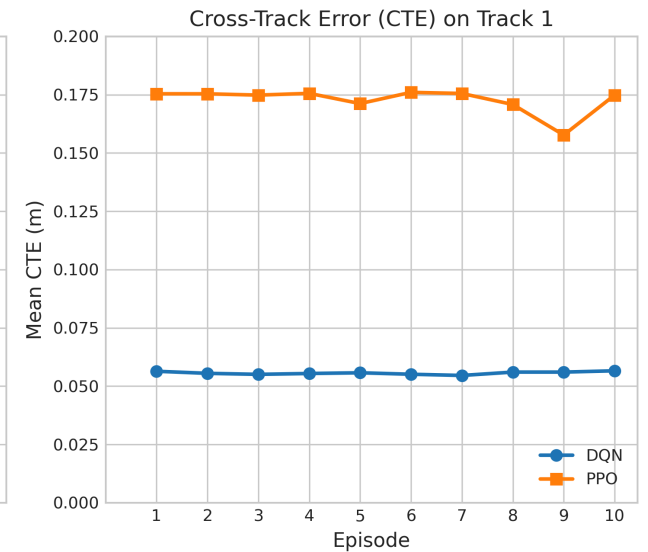
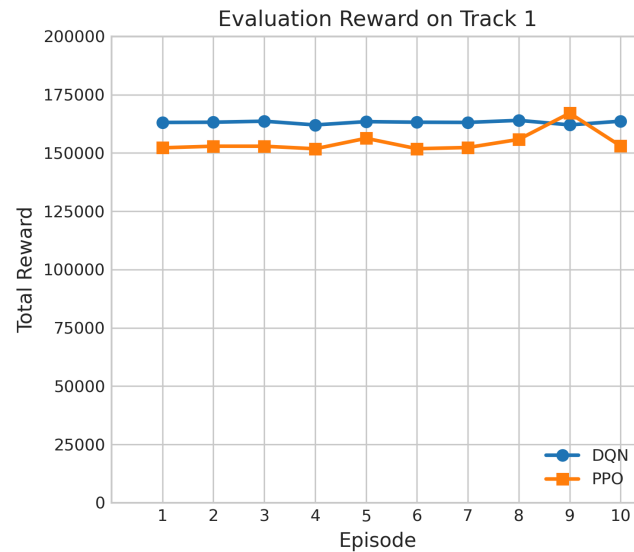
Evaluation:

10 episodes each

Success · CTE · Lap time

Experiment 1: Track 1 results

Metric	PPO T1	DQN T1
Success (%)	100	100
CTE (m)	0.173	0.056
Lap (s)	43.1	49.2
Dist (m)	1209	1074
Near Miss	0.0	0.0



Experiment 2: Cross-Track Generalisation

Goal

Evaluate whether policies trained on Track 1 and fine-tuned on Track 2 can **transfer zero-shot** to the unseen Track 3 without any additional training.

Hypothesis

The alignment-gated reward and multi-sensor features encourage learning structural driving behaviours rather than memorising track-specific layouts.

Training setup

Train on T1
config.yaml

→

Fine-tune on T2
finetune.yaml

→

Zero-shot eval T3
No retraining

Experiment 2: Cross-track generalisation

Metric	PPO T2	DQN T2	PPO T3	DQN T3
Success (%)	100	100	60	0
CTE (m)	0.104	0.060	0.111	0.082
Lap (s)	60.8	74.6	65.1	$\frac{N}{A}$
Dist (m)	1326	1086	1021	742
Near Miss	0.0	0.0	0.2	0.0

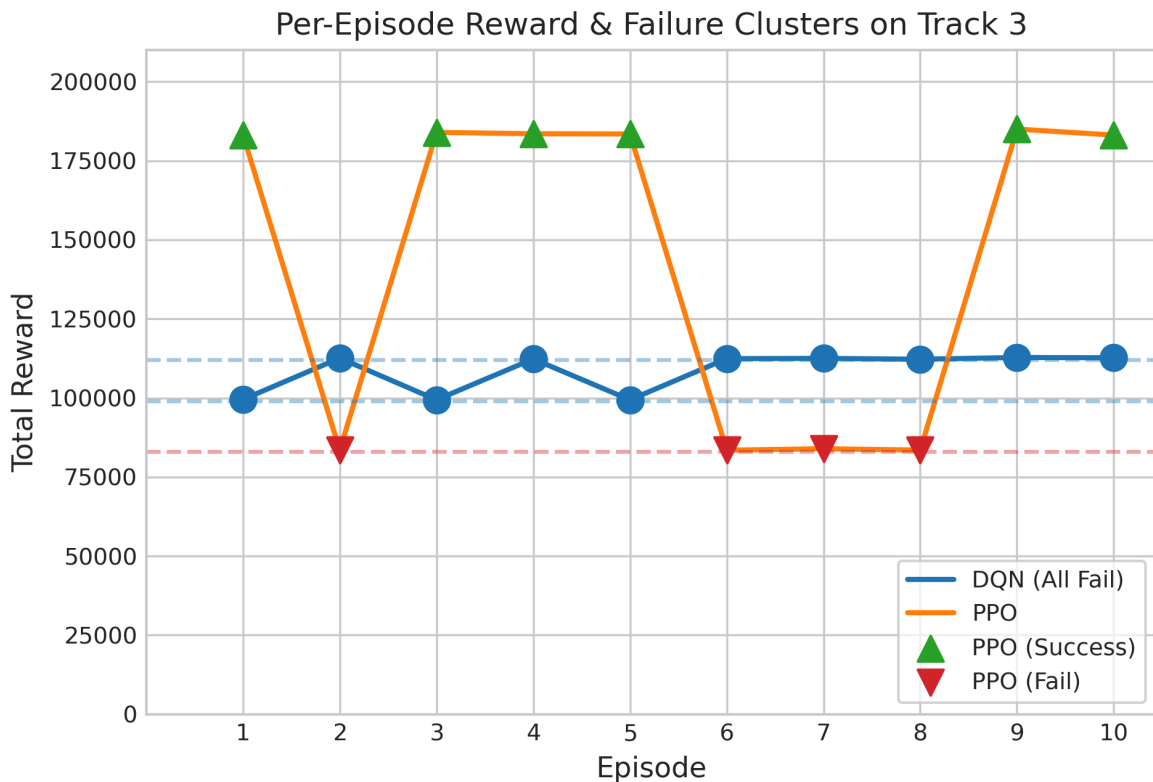
PPO on Track 3 — 60% success

6 / 10 episodes completed. The 4 failures share a common cause: a sharp isolated curve pushes the yellow line out of the camera's lower region-of-interest momentarily, triggering line-lost termination.

DQN on Track 3 — 0% success

All 10 episodes fail at the **same location**: two consecutive turns sharper than anything seen in training. DQN's rigid discrete mapping lacks the fine steering resolution needed to clear tight radii, causing it to run off-track at $\sim 75\%$ completion.

Track 3: Per-Episode Reward & Failure Clusters



Reading the chart

Green triangles mark PPO successes (reward $\approx 180\,000$).

Red triangles mark PPO failures — reward drops to $\approx 85\,000$, always at the same sharp-bend cluster.

DQN (blue) hovers around $100\,000$ — consistent partial completion, but never clears the obstacle section.

PPO's continuous steering gives it **60% generalisation** on unseen geometry; DQN's discrete action mapping is a hard constraint it cannot overcome zero-shot.

Experiment 3: Obstacle Avoidance

Goal

Evaluate the ability of pre-trained agents to **integrate LiDAR distance data** and steer around static barrel obstacles when fine-tuned using `obstacles.yaml` on Track 1 with barrels placed on the road.

Hypothesis

The 1-D LiDAR vector paired with the proximity penalty r_{near} will let agents navigate around static barrels **without losing** baseline lane-keeping competency.

Training setup

Init:
Best baseline
checkpoints
(post T1 + T2)

Fine-tune:
`obstacles.yaml`
Static barrels
on T1

Key metric:
Near Miss count
+
Success rate

Experiment 3: Obstacle avoidance results

Metric	PPO Obs.	DQN Obs.
Success (%)	0	0
CTE (m)	0.153	0.057
Lap (s)	N/A	N/A
Dist (m)	580	120
Near Miss	2.4	1.0

| PPO: Partial avoidance capability

Learned local swerving maneuvers.

When encountering a barrel, **executes a smooth bypass**.

Path corrections push the vehicle into **unrecoverable** states.

Consistently completes $\sim 50\%$ **or** $580m$ of the track.

| DQN: Total failure

Unable to avoid **even a single obstacle**.

Discrete actions cannot blend visual tracking with rapid proximity corrections.

The **dense reward suppresses LiDAR features** entirely to maintain lane position.

The agent **drives straight into barrels**.

Discussion & Conclusion

Key takeaways and future work

Action Space Elasticity: The Key Distinction

| PPO: Continuous control

- **Continuous action** allows discovery of the **balance** between **lane tracking with LiDAR features**.
- **Minute steering** adjustments **preserve line visibility** while initiating a swerve.
- **Partial success**: 60% on unseen geometry, 50% with obstacles.

| DQN: Discrete mapping

- Discrete action requires **large, sudden** steering changes.
- The primary gradient path **suppresses LiDAR features** entirely to maintain tracking stability.
- **Zero success** on sharp unseen curves and total failure on obstacle avoidance.

Key takeaways & future work

Summary of findings

- Both agents achieve **100% success** on simple and medium tracks.
- PPO generalises to unseen geometry at **60%**.
- DQN fails at **0%** due to rigid discrete actions.
- Obstacle avoidance:
 - PPO learns local swerves (**50% track**);
 - DQN ignores obstacles entirely;
- Action-space granularity is a **critical architectural parameter** for multi-modal driving agents.

Future work

1. Network architecture

Explicit attention mechanisms or decoupled value streams to force equal weighting across sensor modalities, preventing feature domination.

2. Reward engineering

Curriculum learning — master obstacle avoidance with LiDAR in isolation **before** reintroducing visual lane tracking data.

3. Extended action spaces

Hierarchical or hybrid discrete-continuous action spaces for DQN.

100%

Track 1 & 2 success

60%

PPO zero-shot T3

0%

DQN zero-shot T3

~ 50%

PPO obstacle track

Demonstration

<https://drive.google.com/drive/folders/1YPLRe94yxvUS1eZuZ-2r8PzcXoAlwRo7?usp=sharing>

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